

Instructor: Date:

Learner: Regional Branch:

Program: Center:

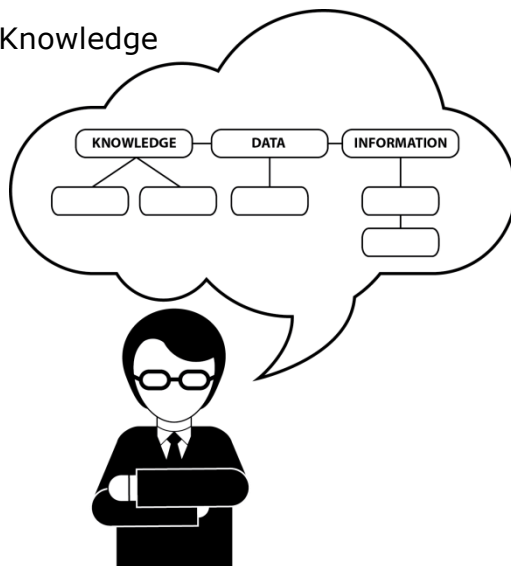
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Data, Information and Knowledge



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- How do we store information we get about the world?
- Why is it important to keep records of general knowledge and people?

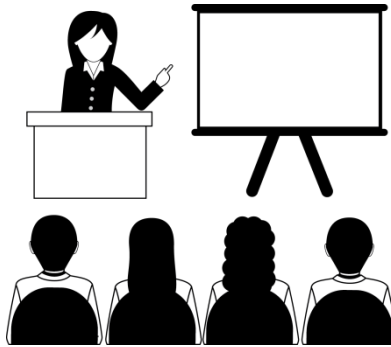
Can Do Statements:

- I can read and write about data, information and knowledge.
- I can compare data, information and knowledge.
- I can write meaningful sentences to explain the concepts of data, information and knowledge.



Let's Explore!

What is the difference between data, information and knowledge?



Instructions: Read the text below.

- Teacher:** Ok class, we covered an interesting topic last class. Who can remind me what we studied yesterday? How about you Victor?
- Victor:** We worked on the difference between data, information, and knowledge. You told us that we could mix up the three concepts. So, we should be careful when defining each one of them. First, we discussed knowledge. Simply put, knowledge is what we know and is stored in our brains, like a map of the world. Currently, you can only keep knowledge in a human brain. Computers are not artificial brains because they don't understand what they process and can't make independent decisions. Thus, internet is not a network of human knowledge. The brain uses two sources to build knowledge: data and information.
- Teacher:** Excellent, Victor. Now, Cristina could you tell me what data is?
- Cristina:** Data are the facts in reality. For example, your age or your height. Data is a description of the real world as we perceive it with our senses. In the past, humans could only build knowledge of the world by direct experience. Before they started storing data as information, you couldn't know a person's height by looking at a database; you had to measure or ask the person directly. Nowadays, we can store it in computers and check it whenever we want.
- Teacher:** Great job Cristina. Finally, Victor can you tell me what information is?
- Victor:** Information is how humans expand their knowledge beyond the data collected by their senses. In other words, information is the record of data we have of the world. For example, a photograph of you is information on how you look, but your appearance in the real world is data. You can send, transform or even destroy the photograph, but your looks, in reality, are not affected.
- Teacher:** Very good Victor. OK class; remember you shouldn't confuse the concepts of data, knowledge, and information. For next class, bring some ideas about how these concepts have changed, in the past, we had a different concept of information and data, these days, data, information and knowledge are related one to each other.



Let's Practice!

Victor remembers everything about last class.

Instructions: Circle the correct option to complete the sentence. Read the sentences aloud. Check your pronunciation with your instructor.

1. Yesterday, we (work / worked) on the difference between data, information and knowledge. Today we have the exam.

2. Last class, you (told / tell) us that we could mix up the three concepts, but we have them clear now.

3. Nowadays, we (know / knew) that you can only keep knowledge in a human brain which uses two sources to build knowledge. In the past people only knew what they could perceive with their senses.



Let's Practice!

It's important to know the difference between these three concepts.

Instructions: Read the sentences. Write the missing word in the blank. Use: You should learn, Could you tell, I can explain. Practice saying the sentences aloud. Check your pronunciation with your instructor.

1. _____ what knowledge is: knowledge is what we know and is stored in our brains, like a map of the world.
2. _____ me what data is? Data is a description of the real world as perceived by our senses.
3. _____ the concept of information: it is how humans expand their knowledge beyond the data collected by their senses.



Extension Activity

Let's study the difference between the concepts.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more evidence about the difference between data, information and knowledge.
- Write the important differences in a comparative chart.
- Share the comparative chart with your partners.

Self-Assessment

Yes

No

- I look for evidence related with data, information and knowledge.
- I write about data, information and knowledge in a comparative chart.
- I share information with my partners.

☐ ☐☐ ☐☐ ☐

Target Words

- **sense** (noun): any of the human faculties to perceive reality
e.g. Data is a description of the real world as we perceive it with our senses.
- **beyond** (preposition): outside the limits of something
e.g. Information is how humans expand their knowledge beyond the data collected by their senses.
- **looks** (noun): aspect or appearance
e.g. You can send, transform or even destroy the photograph, but your looks in reality are not affected.
- **simply put** (expression): in clear terms
e.g. Simply put, knowledge is what we know and is stored in our brains.
- **independent** (adjective): without any intervention or help
e.g. Computers can't make independent decisions.
- **height** (noun): body extent measured from lowest to highest point
e.g. Data are the facts in reality, for example your age or your height.



Learning Points

Simple Past vs. Simple Present

We can use the **simple past** tense and the **simple present** tense to describe actions we finished in the past or actions we routinely do.

We use the simple past tense when we want to describe an activity that we started and finished at some point in the past e.g. last Monday, yesterday, in 2010, etc. In other words, there is not a direct connection to the present time when we are talking.

Example:

*Yesterday in class, we **studied** the difference between knowledge, data and information.*

On the other hand, we use the present simple tense when we want to describe general or habitual activities that are repeated regularly e.g. every day, on weekends, every Tuesday, etc. Generally, the simple present tense refers to habits or routines.

Example:

*Every semester, I **watch** the movie "Inside Out" with my students. It **explains** in a friendly manner some complicated topics like memory and behavior.*

We can use these two tenses in a sentence to show contrast between the past and the present.

Example:

In the past, people **didn't have** to worry about moral standards in technology, nowadays, people **exceed** the limits of life using it.

Examples

- *In the past, we had a different concept of information and data, these days, data, information and knowledge are related one to each other.*
- *In the past, humans could only build knowledge of the world by direct experience. Before they started storing data as information, you couldn't know a person's height by looking at a database; you had to measure or ask the person directly. Nowadays, we can stored it in computes and check it whenever we want.*





Learning Points

Modal verbs can, could and should (review)

We use the modal verbs **can, could and should** for expressing abilities and recommendations.

We use can and could to describe skills or possibilities. In the present tense we use can, but in the past tense we use could. Also, could can be used for requests and suggestions.

Examples:

Skills

They **can** store data, but they **can't** store knowledge.

She **could** store data, but she **couldn't** store knowledge.

Possibilities

We **could** understand better the difference if we had paid attention.

Requests

Could/Can you explain me the difference between knowledge, data and information?

On the other hand, we use should to give advice. It expresses a mild obligation or moral responsibility.

Examples:

People **should** store their data in safer ways.

You **shouldn't** share your data with everyone.

Examples

- *You can only keep knowledge in a human brain.*
- *We should be careful when defining each one of them.*
- *In the past humans could only build knowledge of the world by direct experience.*